SUPPLEMENTARY MATERIALS: When Can Markets Crash Without News?

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SM1. Scope of the supplement. These panels and the earlier nonoptimal two-day witness are preserved for replication and illustration. They are not independent estimates of real market risk. All mathematical claims and the improved interval-certified construction appear in the main manuscript.

SM2. Earlier exploratory two-day execution. The preceding theorems form a hierarchy of *necessary* tests. Passing the rolling-band bound, the scalar selling-rate test, the dollar-closure preconditioning test, and the cut-capacity test still does not prove a path exists. We therefore turn from exclusion and lower bounds to a constructive example that enforces the constraints *simultaneously*. To test repeated *partial-coverage* lower-limit closes, we constructed a continuous $N = 10$, $k = 9$ example in the $g(S) \equiv \eta$ market. The nine targets use symmetric $\pm 3\%$ proportional bands; the compensator uses a wider symmetric $\pm 25\%$ band. These are *hypothetical normalized assets*, not calibrated TSE stocks. Set $F = \eta = 1$, $\rho = 0.4$, equal session lengths $\tau_1 = \tau_2 = 20$, and a common gross share-rate limit $\bar{v} = 1$.

Write $S_1 = \cdots = S_9 = x(t)$ and $S_{10} = y(t)$. On each session let $x(t)$ be a continuous piecewise-linear path: it rises to a first-session plateau h_1 then closes at 0.97, rises to a second-session plateau $h_2 = 0.995$ then closes at $0.97^2 = 0.9409$. All ramps last 0.7 trading-time units. Dollar pairing uniquely determines the compensator's ODE via

$$u(t) = x(t)[\dot{x}(t) + \rho(x(t) - 1)], \qquad \dot{y} = -\rho(y - 1) - \frac{9u(t)}{y(t)}, \quad y(0) = 1. \tag{SM2.1}$$

The first plateau $h_1 \approx 1.0142982125$ is selected from the *exact piecewise-polynomial* equation

$$X_i(T_2) = \frac{x(T_2)^2 - 1}{2} + \rho \int_0^{T_2} x(t)[x(t) - 1]\,\mathrm{d}t = 0, \quad i = 1, \ldots, 9. \tag{SM2.2}$$

Because $\dot{X}_{10} = -9\dot{X}_1$, the compensator's terminal executed-dollar balance closes as well. Independent high-accuracy ODE and Runge–Kutta integrations agree for $y(T_2)$ to approximately 2×10^{-12}; band and rate slack were checked on a dense time grid. These numerical checks are an explicit *witness*, not a proof of general N-asset sufficiency or a certified interval-enclosure theorem.

Figure SM1 displays the earlier exploratory two-session price paths.

Figure SM2 displays the ledgers and rates of that exploratory witness.

The example proves neither a trader's incentive to undertake the sequence nor a *locked* limit-down order book. It does show numerically that rolling bands, finite executed-share rates, no fundamental-news shocks, and terminal dollar closure can be compatible with *multiple* lower-limit closes in a broad subset of assets. For a crisis threshold, the breadth requirement

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Table SM1

Two-session $N = 10$ deterministic closing-limit witness (normalized units). Day 1 and Day 2 target losses are 3% each relative to the reset reference; the terminal systemic functional is smaller because the compensator rises.

Quantity	Verified value
Nine target closing prices, session 1 / session 2	0.970000 / 0.940900
Compensator closing prices, session 1 / session 2	1.228104 / 1.417574
Maximum individual gross executed-share rate	$0.680780 < \bar{v} = 1$
Target executed-dollar balance, day 1 / day 2	$+0.078235$ / -0.078235
Maximum absolute terminal net-dollar balance	$< 7 \times 10^{-15}$
Total paired-dollar turnover / largest gross share sales	2.59310 / 1.46241
Terminal $\mathcal{C}_{0.9}(T_2)$ / breadth loss $Q_{0.9}(T_2)$	0.019932 / 0.060918

alone does not suffice: for this path $Q_{0.9}(T_2) = 0.060918$ but the average-loss restriction binds, giving $\mathcal{C}_{0.9}(T_2) = 0.019932$ (thus, for example, $\delta = 0.015$ is reached). No claim of an exact minimum rate, minimum turnover, or equilibrium occurrence follows.

SM3. Other diagnostic displays. Figures SM3 and SM4 present the market primitives and local elasticity examples, respectively.

Figures SM5 and SM6 illustrate the model-specific full-coverage boundary. Figures SM7 and SM8 illustrate the necessary resource bounds and their asset-count dependence.

Figure SM9 displays the bounded-rate continuous example and its closed dollar ledgers. Figure SM10 compares the rate and price-limit exclusion bounds. Figures SM11 and SM12 show the scalar necessary-rate floor and the region covered by the all-asset obstruction.

Figure SM13 distinguishes the necessary rate floor from the certified feasible rate; Figure SM14 displays the funding-capacity constraint as the falling majority grows.

Figure SM15 displays terminally dollar-closed price paths, while Figure SM16 distinguishes dollar closure from net share positions. Figure SM17 illustrates the construction families with bounded feasible resources. Figure SM18 shows why the earlier pulse example does not satisfy the newer execution restrictions.

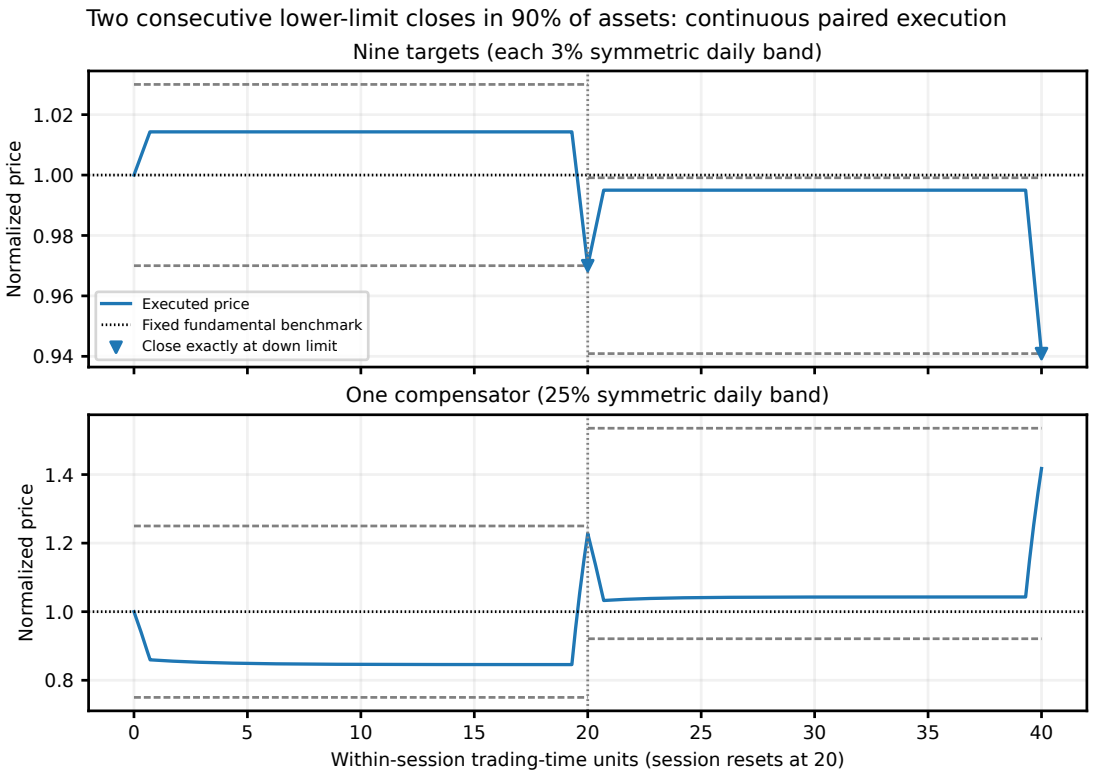


Figure SM1. ***Repeated partial-coverage lower-limit closing prices under rolling daily bands.*** *The same nine target assets close at their 3% lower limits twice. The tenth, higher-band compensating asset stays within its own 25% limits. The displayed paths contain no instantaneous trades. The horizon and bands are specified model inputs, not observations.*

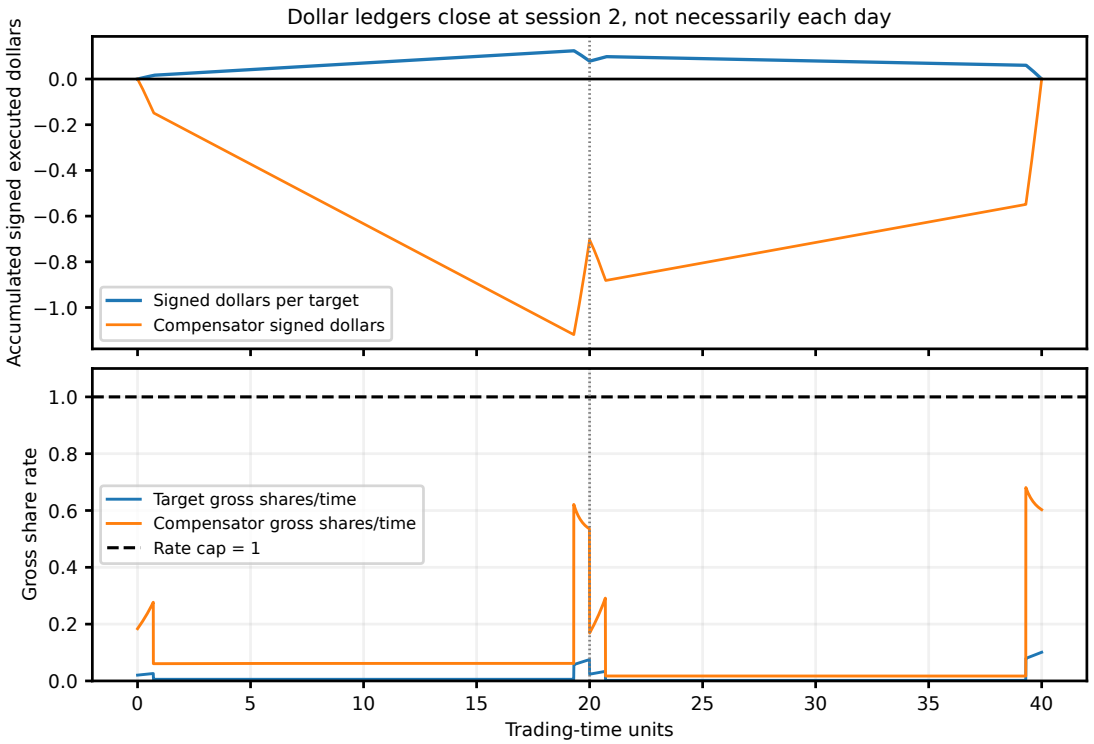
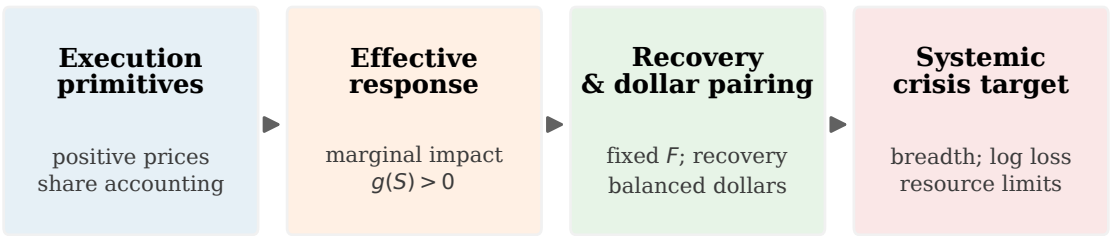


Figure SM2. ***Trading ledgers and gross-rate check for the same two-session witness.*** *The first session builds positive net dollar flow in the targets; the second reverses it, leaving every asset's final signed-dollar balance zero. Gross rates remain below the imposed share-rate limit; dollar balance is enforced continuously, not just at maturity.*



Price priority and accounting are primitive; impact and resilience are imposed model conditions

Figure SM3. ***One explicitly defined ideal market.*** *Price priority, signed-share accounting, effective price impact, resilience, and paired-dollar trading have distinct mathematical roles. All quantitative panels in this paper are illustrations of this specified model rather than observations of a live order book.*

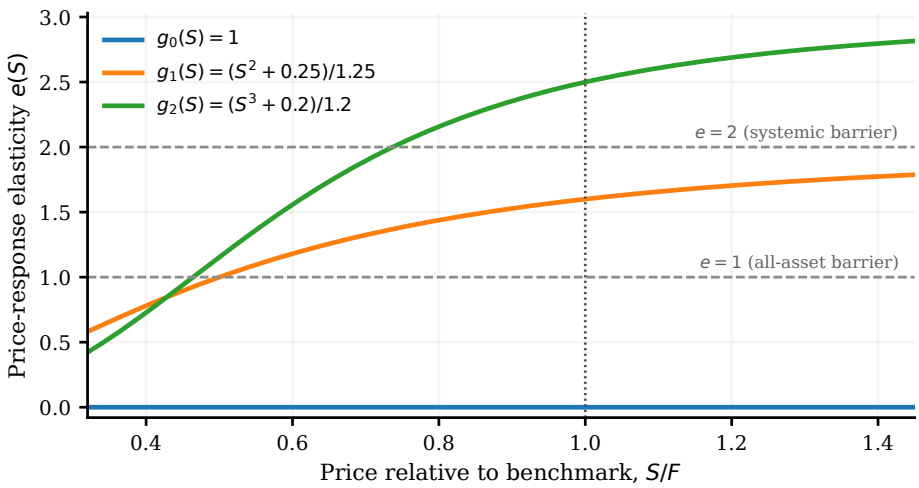


Figure SM4. ***Local elasticity of three positive, smooth, non-power-law impact functions.*** *The dotted vertical line marks the common initial price $S = F$. The global elasticity-barrier theorems in the main manuscript require their elasticity inequalities throughout the* relevant attainable price region*; a value at F alone gives no global classification. This is a formula-based illustration, not empirical market depth.*

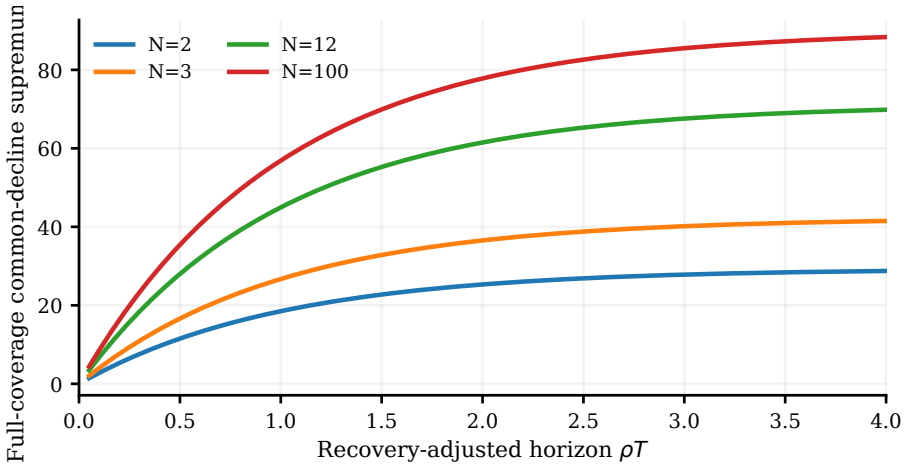


Figure SM5. ***A sharp, model-specific common-decline boundary.*** *For $g = \eta$ and $k = N$, the upper envelope of the common simple-price decline is $100(1 - e^{-\delta_{\mathrm{crit}}})\%$. Every strictly smaller log-loss threshold is reachable with some finite budget; the boundary itself is not attained with strictly positive prices. This is not a probability curve.*

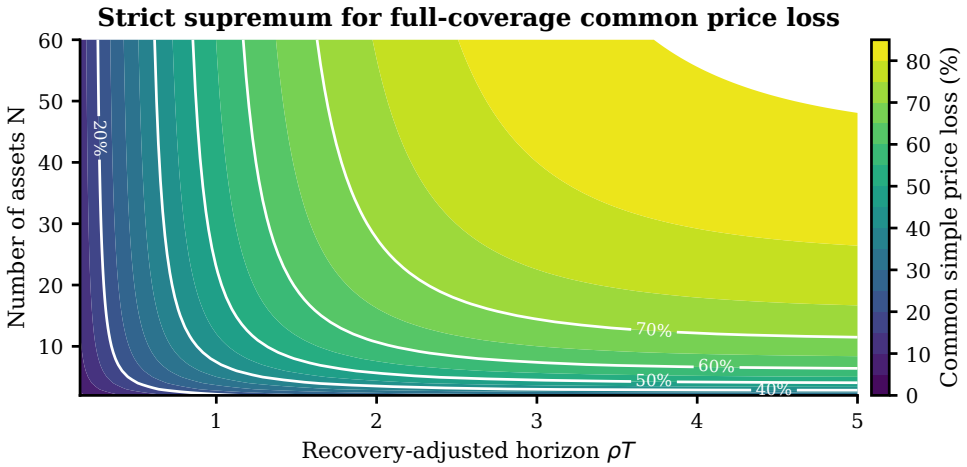


Figure SM6. ***The full-coverage boundary across asset count and recovery time.*** *Color reports* $100(1 - e^{-\delta_{\mathrm{crit}}(N,T)})$*, the supremum of simultaneous* simple-price *losses for constant per-share impact with individually closed terminal dollar flows. No probability, observed drawdown, or finite-budget optimum is inferred.*

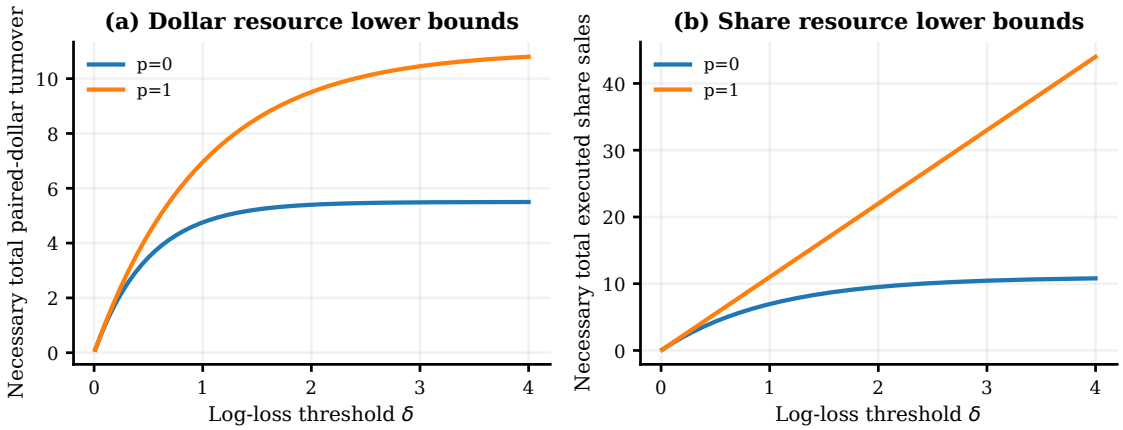


Figure SM7. ***General necessary resource floors illustrated for the power-response family.*** *With* $F = \eta = 1$ *and* $k = 11$*, the curves evaluate* $k \int_{e^{-\delta}}^{1} S/g(S)\,\mathrm{d}S$ *(dollars) and* $k \int_{e^{-\delta}}^{1} 1/g(S)\,\mathrm{d}S$ *(shares), for* $p = 0, 1$*. The bounds are* not *the optimal execution cost and do not establish reachability.*

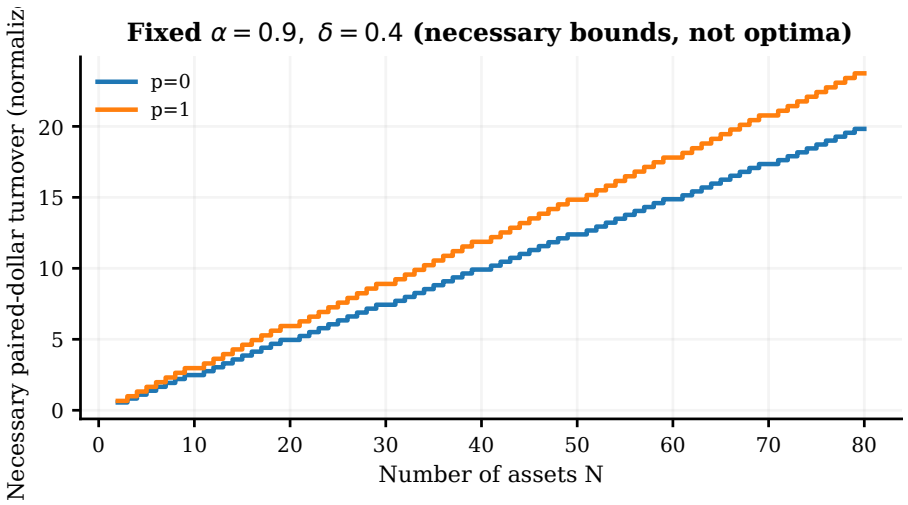


Figure SM8. ***Dependence on asset number of the necessary dollar-resource bound.*** *The experiment copies per-asset benchmark price and response parameters, fixes $\alpha = 0.9$ and $\delta = 0.4$, and increases the number of assets N. The steps arise from the discrete coverage count $\lceil \alpha N \rceil$. It is not a capitalization-scaled prediction for an actual stock index.*

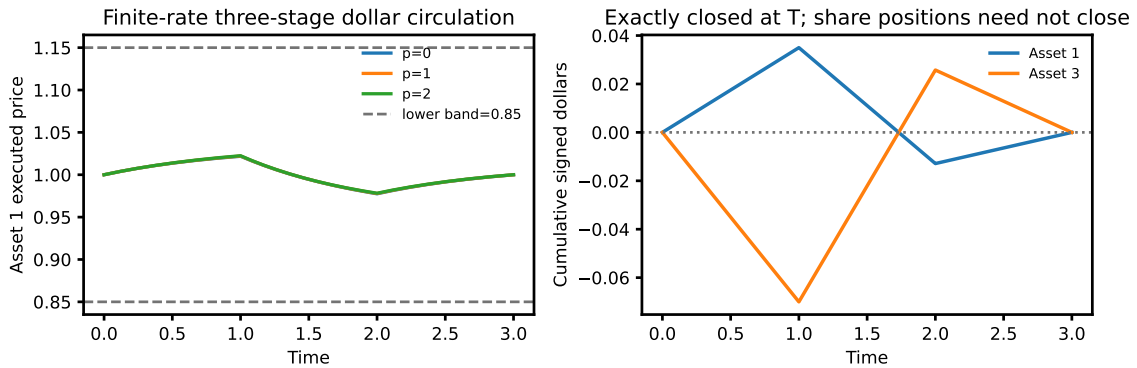


Figure SM9. ***A new bounded-rate, terminally dollar-closed path (not a pulse approximation).*** *For $N = 3$, $T = 3$, $a = .035$, $F = \rho = \eta = 1$, and $u = (1, 1, -2)$, the price trajectories under $p = 0, 1, 2$ share the same absolutely continuous dollar-rate signal; right panel shows dollar ledgers that close exactly. The peak per-asset gross share rate is below 0.15, and all prices remain inside a $\pm 15\%$ band.*

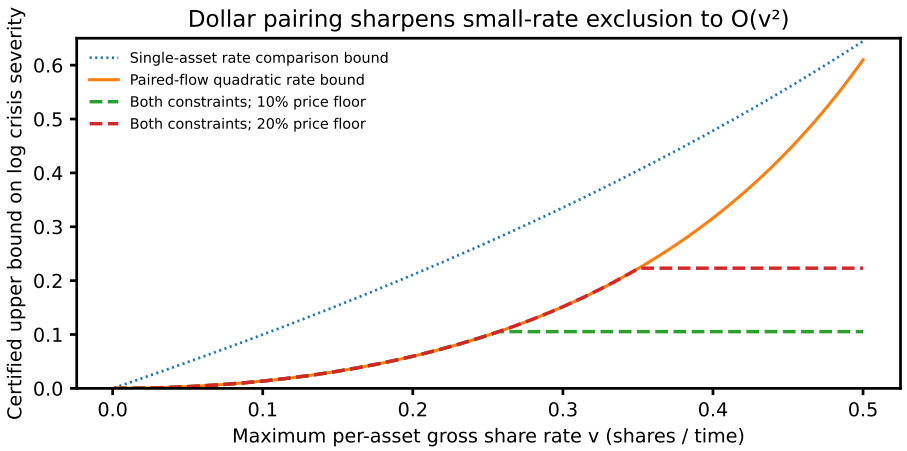


Figure SM10. ***Exclusion certificates, not exact reachable maxima.*** *For $g = \eta = F = \rho = 1$ and $T = 3$, a paired-flow quadratic estimate improves the single-price comparison at small rates; a single-session lower price band provides an additional hard ceiling. Neither curve reports optimal crisis control or an observed market frequency.*

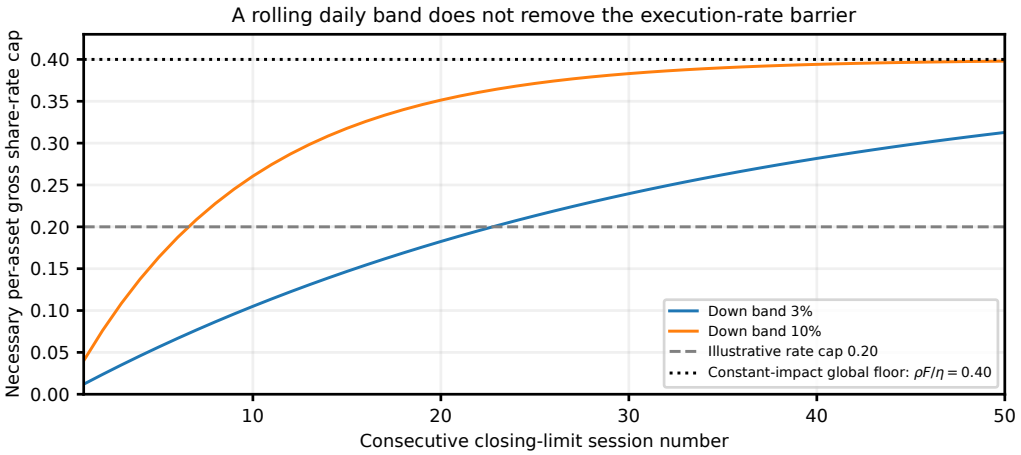


Figure SM11. ***Necessary gross share-rate floor for repeated lower-limit closes.*** *The scalar constant-impact necessary-rate formula in the main manuscript in the scalar constant-impact comparator, $F = \eta = 1$, $\rho = 0.4$, $\tau = 20$. These are* necessary *per-asset rates; paired execution, available inventory, and the compensator's own daily band can make the actual market threshold higher.*

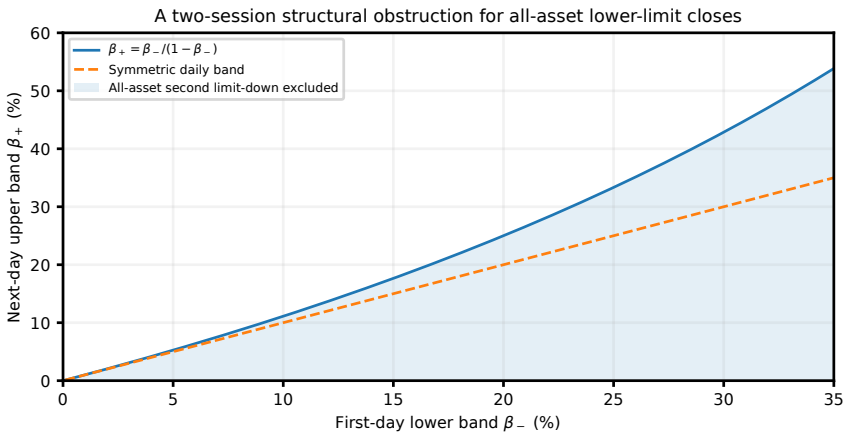


Figure SM12. ***Parameter region in which the second all-asset lower-limit close is ruled out.*** *The region beneath the curve satisfies* $(1+\beta_+)(1-\beta_-) \le 1$. *Symmetric proportional bands lie strictly inside this region. This is a conditional invariant-region result, not a model of exchange-wide order queues.*

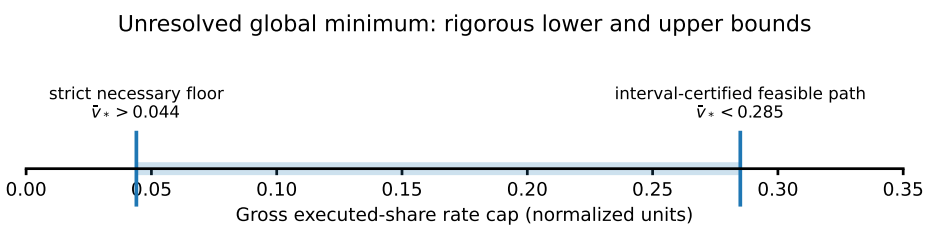


Figure SM13. ***What is and is not optimized.*** *A strict rate floor from the existing quadratic systemic bound and a certified feasible execution path bracket the unknown global minimum* $\bar{v}_*$. *The interval is a logical admissibility bracket, not a statistical confidence interval.*

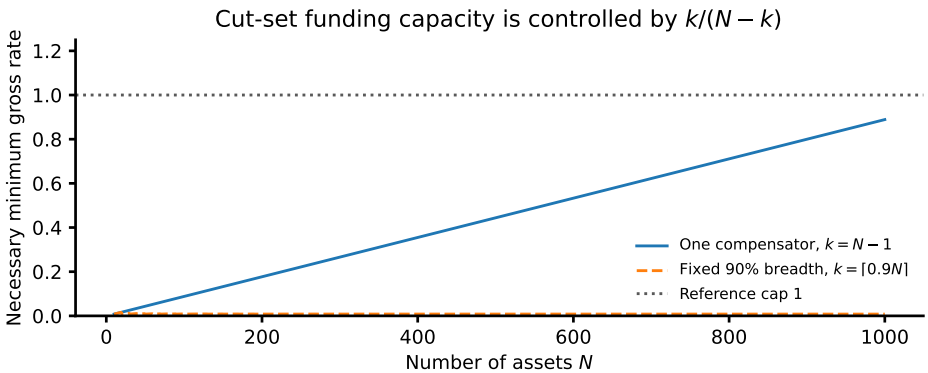


Figure SM14. ***A funding-capacity constraint that grows with the falling majority.*** *The necessary second-day rate in the main manuscript increases as* $k/(N-k)$ *when a single asset absorbs the net matched dollars of an expanding majority. This does not prove that a feasible market exists at every plotted* N.

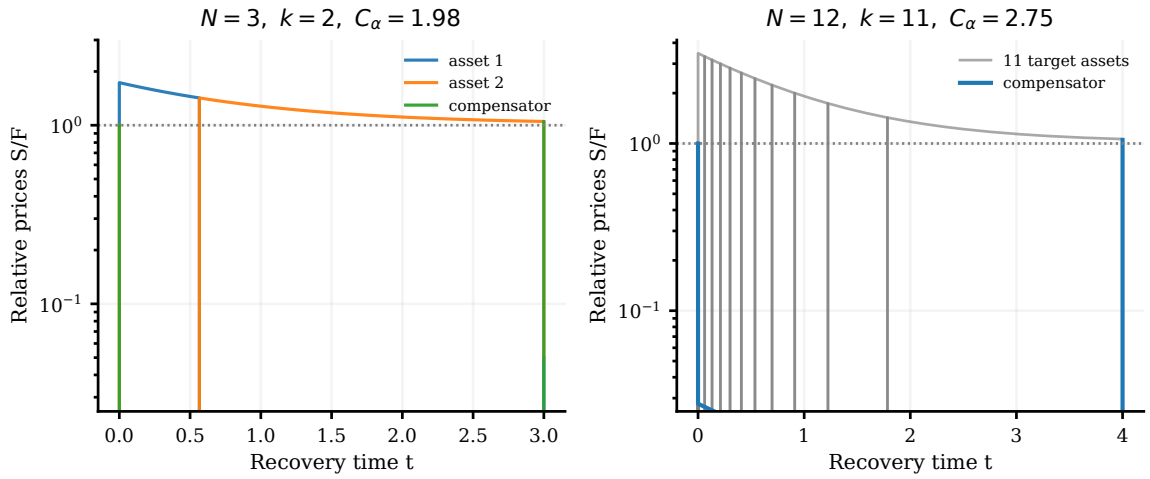


Figure SM15. ***Constructed price paths with terminally closed dollar flows.*** *For $N = 3$ and $N = 12$, one compensating asset ends above its initial value, while the other $N - 1$ prices fall by the common target log loss $L = 3$. Logarithmic vertical scales keep near-zero prices visible. Kinks and jumps are permitted control pulses, not simulated exchange events.*

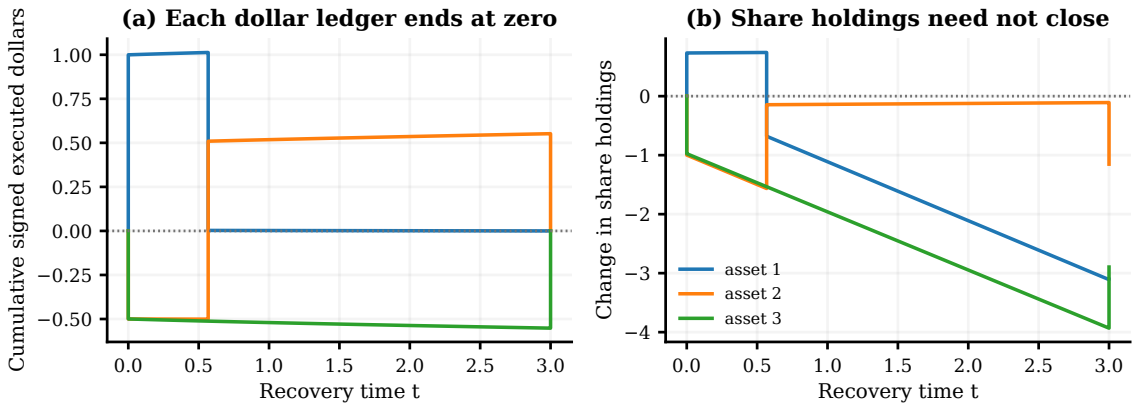


Figure SM16. ***Net executed dollars are not net shares.*** *A separate $N = 3$, $k = 2$, $T = 3$, $L = 5$ witness has $X_i(T) = 0$ for every asset (left), but its net share positions generally differ from zero (right). Initial share inventories must cover gross sales; transaction closure alone does not supply them.*

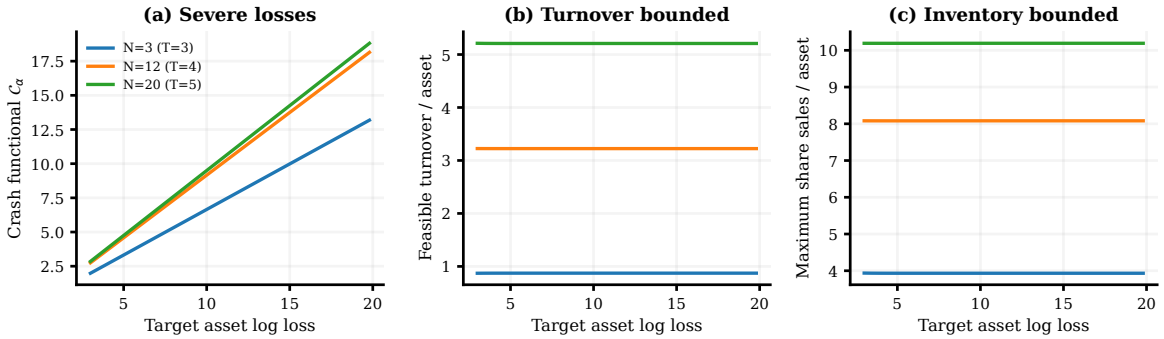


Figure SM17. ***Increasing crisis severity with bounded* feasible *resources.*** *Selected $N = 3, 12, 20$ construction families use $k = N - 1$ and horizons $T = 3, 4, 5$, respectively. As the terminal target-price log loss increases, achieved $\mathcal{C}_\alpha$ rises (left), whereas the particular paths' turnover per asset (center) and largest cumulative share sale (right) remain bounded. The different horizons and prices are model choices, not a cross-market calibration or an optimum.*

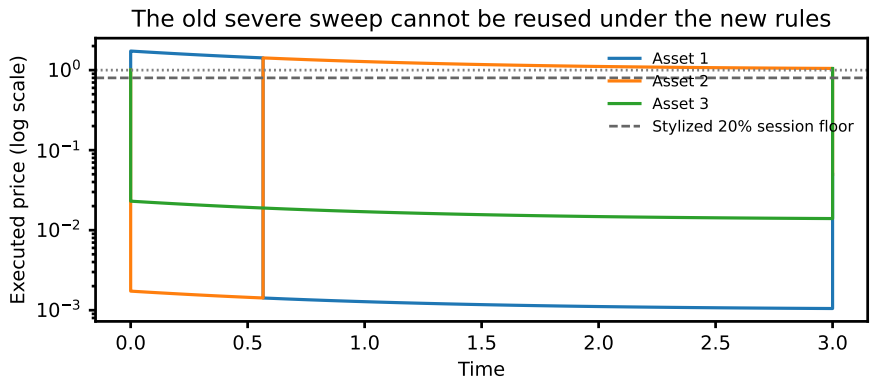


Figure SM18. ***Diagnostic rejection of an old severe path under new execution restrictions.*** *The $N = 3$, $p = 0$ severe-dollar-closed witness from the unrestricted model crosses a stylized one-session 20% lower band and includes nonzero instantaneous pulses. Its exact closed dollar ledger therefore* does not *certify reachability under either a finite gross rate or the price-limit protocol.*